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The hybrid renewable energy community approach (HyRECA): Synergising electricity access with bush encroachment mitigation in rural Southern Africa

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ABSTRACT

Rural electrification and bush encroachment in Southern Africa have long been studied separately. Research shows that hybrid renewable energy systems (HRES) offer viable solutions for off-grid communities, and bush encroachment mitigation is widely recognised as an ecological priority. Using Jamataka, Botswana, as a case study, this study integrates both, asking under which conditions encroacher bush-based HRES are economically viable, which electricity access levels local populations can afford, and whether biomass utilisation contributes to mitigation. Findings indicate strong potential for bush-based HRES in off-grid areas, but competitiveness declines where low-cost grid electricity is available. Economic viability is examined through levelised cost of electricity (LCOE) across alternative HRES configurations. HOMER Pro® simulations indicate PV/biomass/battery systems are most cost-effective off-grid, achieving 100 % renewable fractions, near zero CO₂ emissions, and LCOE of 0.246 \$/kWh. In grid-connected contexts without feed-in, PV/diesel/grid combinations dominate (LCOE 0.117 \$/kWh) due to low tariffs, but with higher emissions. A novel affordability analysis using household willingness to pay and the World Bank Multi-Tier Framework indicates that all households reach at least Tier 3 electricity access under grid-connected systems and >70 % off-grid, supporting household-use of medium-power appliances. Sustainable biomass extraction clears ~12 ha annually (~120 t) in Jamataka, aiding bush encroachment mitigation. Upscaling indicates potential to electrify 1.35 million people in >900 bush-encroached villages across Botswana, Namibia, and South Africa, using <1 % of encroached land – highlighting significant untapped potential. Future research should validate findings through geospatial modelling with high-resolution biomass data and spatially sensitive biomass cost models for region-wide planning.

1. Introduction

Recent political commitments call for holistic solutions to accelerate Africa's green transition and electricity access. Globally, the United Nation's SDG 7 targets affordable, reliable, sustainable, and modern energy for all, noting that Africa accounts for 80 % of those lacking access [1]. The African Union's Agenda 2063 promotes climate-resilient communities [2], while Southern African governments emphasise local

value chains and economic transformation, as seen in Botswana's National Mindset Change Campaign [3]. This study addresses two key challenges that are specific to Southern Africa and align with these political commitments.

The first challenge is **access to clean, affordable, and reliable energy**. Botswana, Namibia, and South Africa together have 2.7 million rural residents without electricity access [4]. Grid extension is costly and difficult due to low population density, while national plans often prioritise large-scale coal, gas, or hydropower projects that poorly address

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Nomenclature

\$	US Dollar
$A_{bush, total}$	long term useable area as bush feedstock
D_{bush}	bush density in mass per area
$d_{bush, road}$	specific long term bush mass per road length
E_{el}	Electrical energy output
LHV_{SG}	Lower heating value of SG
l_{road}	long term useable length of public roads as bush feedstock
m_{bush}	bush mass
$r_{regrowth}$	Bush complete regeneration period after harvesting
$w_{roadside}$	width safety strip besides public roads
$\eta_{el, SG}$	Electrical efficiency of SG
η_{GR}	Gasification ratio reactor, biomass to syngas (mass-related)

Abbreviations

AgriPV	Agriculture-integrated photovoltaics
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BESS	Battery energy storage system
BWP	Botswana Pula
CAPEX	Capital expenditure
DG	Diesel Generator
DESCO	Distributed Energy Services Company
HRES	Hybrid renewable energy system
HyRECA	Hybrid Renewable Energy Community Approach
JMS	Jamataka Multipurpose Co-operative Society Limited
LCOE	Levelised cost of electricity
O&M	Operations and maintenance
PV	Solar photovoltaics
SADC	Southern African Development Community
SDG	Sustainable Development Goal
SG	Syngas-based power generation system
WTP	Willingness to pay

rural needs [5]. Existing grids are highly unreliable, with frequent load shedding and nationwide outages disrupting households, businesses, and public services, as notably observed in Botswana in 2025 [6]. Despite their suitability, community-based renewable energy projects remain scarce, are often designed top-down and depend on goodwill of international organisations [5].

Another significant challenge faced by rural communities in Southern Africa is **bush encroachment**, which reduces land productivity, limits water availability, increases fire risks, and negatively impacts biodiversity, thereby constraining agricultural and pastoral activities in semi-arid regions reliant on rainfed subsistence farming [7–12]. Despite control efforts, resources remain insufficient, and the problem persists across large areas of Botswana, Namibia, and South Africa.

In response to these two challenges in Southern Africa – (i) **access to clean, affordable, and reliable energy** and (ii) **bush encroachment** – the Hybrid Renewable Energy Community Approach (HyRECA) innovates small-scale, off-grid capable hybrid renewable energy systems (HRES) based on agriculture-integrated photovoltaics (AgriPV) and the utilisation of abundant waste biomass from encroacher bush in syngas-based power generation systems (SG). The study explores: (i) economic feasibility of bush-based HRES via LCOE; (ii) affordable electricity tiers based on willingness to pay (WTP); and (iii) bush mitigation potential through biomass use and cleared land area. [Section 2](#) reviews literature covering HRES relying on SGs and bush encroachment ecology – identifying this study’s research gap and scope of work. [Section 3](#) presents HyRECA’s technical and socio-economic framework, followed by [Section 4](#) detailing the methodology, including case study, simulation setup, impact assessment, and transnational upscaling. [Section 5](#) reports and discusses results, while [Section 6](#) concludes with key findings, limitations, and recommendations for future research and policy insights.

2. Literature

This chapter reviews HRES, focusing on SG-based approaches, and examines the ecology of bush encroachment in Southern Africa to identify research gaps and define the study’s scope.

2.1. Hybrid energy solutions

A key challenge in off-grid electrification is supply unpredictability when relying on a single energy source. For example, solar photovoltaic (PV) without battery energy storage system (BESS) cannot consistently meet electricity demand, as night-time or overcast conditions reduce yields and cause interruptions [13]. Combining local resources such as

solar and biomass in hybrid systems offers greater resilience and enables 24/7 electricity access where grid extension is prohibitively expensive [10]. The full review methodology and more information are found in supplementary materials S1 & S2.

Recent literature (2024–2025) confirms PV as the dominant renewable technology in nearly all hybrid systems, often combined with wind turbines and BESS for short-term balancing. Dispatchable generation typically relies on diesel or biomass, with biogas being most common, and SGs still rare. Median levelised cost of electricity (LCOE) is lowest for grid-connected systems (0.07 \$₂₀₂₅/kWh) and HRES’ with micro-hydro (0.152 \$₂₀₂₅/kWh), followed by biogas (0.178 \$₂₀₂₅/kWh). HRES based on SG (0.229 \$₂₀₂₅/kWh) and diesel generators (DG) (0.221 \$₂₀₂₅/kWh) are costliest, though SG offer potential advantages when low-cost biomass residues are locally available.

An extended review identifies 17 syngas-based case studies (2012–2025), mostly optimised with HOMER Pro® (12). The dominant configuration is PV/SG (11), with a few multi-technology hybrids and two non-hybrid systems. BESS appear in 14 cases and two assume grid-connection [14,15]. Most studies are from India (11), followed by Bangladesh (2), Cameroon (2), and one each in Nigeria and Liberia. Feedstocks are mainly agricultural residues (e.g., rice husk, straw, coconut) and wood waste, with prices highly site-dependent (median: 42 \$₂₀₂₅/t, range: 16–67 \$₂₀₂₅/t). SGs show promise where low-cost residues are available, but high capital expenditure (CAPEX) and location-specific biomass constraints limit their competitiveness.

Three studies compare off-grid with grid-connected systems, all finding grid-connected options cheaper than off-grid SG-based HRES [14–16]. Islam et al. (2018) examine consumer electricity tariffs, government subsidies and financing models for the off-grid system to reach grid parity [14]. While [14,16] assume a fully reliable grid, Rajbongshi et al. (2017) contrasts this with scenarios of limited availability (6–24 h/day) [15]. However, older HOMER Pro® versions modelled outages as fixed schedules; the updated version enables randomised outage simulation, which this study uses to address a gap in reliability modelling.

Affordability aspects are rarely considered quantitatively. Singh et al. (2019), e.g., complement their HOMER Pro® analysis with a business model review and recommendations, noting that their calculated LCOE exceeds what poor households can currently afford [17]. Only three of the 17 reviewed studies note that LCOE alone does not reflect affordability. Harris et al. (2017) address this by introducing an optimisation method focused on minimising household payments [18]. A less fundamental approach is applied by Eziyi et al. (2014): the authors use a community’s average ability to pay to compare with the calculated LCOE [19]. While this approach provides a first-level estimate, it overlooks wealth disparities within the community. Few other

examples exist beyond the scope of this review: Similarly to [19], Alfaro & Miller (2014) adopt a single literature-derived WTP value for Liberia [20], whereas Indrawan et al. (2020) use national survey data to define ten income-based WTP levels for two Indonesian cities. Their analysis estimates rooftop PV viability by identifying the break-even year between declining LCOE and income-based WTP [21]. Unlike this study's rural electrification focus, their context differs substantially and does not address electricity access tiers based on household WTP – highlighting a gap this work aims to fill.

2.2. Bush encroachment in Southern Africa

Bush encroachment is the spread of indigenous woody species into treeless ecosystems such as grasslands, savannas, and fynbos [22,23]. Bush encroachment is defined as the abnormal increase in native bush cover and intensity. Encroacher species occur across climatic zones, with notable examples including *Prosopis* in Namibia [24] and *Acacia tortilis* in Botswana [25]. Expansion of bush vegetation is partly caused by climate change through the “CO₂ fertilisation effect”, which accelerates growth, and by past land uses such as overgrazing that give woody species a competitive edge [26]. Key impacts include declining land productivity, biodiversity erosion, increased fire hazards and reduced water availability [22,27–29]. While some argue that such expansions could be beneficial from a carbon sequestration perspective, this assumption is disputed due to complex soil processes and highly variable carbon balances influenced by local conditions like rainfall and soil type [30,31].

The consensus is that bush encroachment in Southern Africa must be controlled if not eradicated. Existing programmes lack sufficient resources and biomass resulting from land clearing operations is usually wasted, either burned on-site or left to decay [32,33]. Yet, controlling these invasions aligns well with growing biomass demand for energy and other products. Realising this potential depends on several factors such as the economics of related value chains (e.g. access to and delivery of the biomass at competitive costs) and the availability of sufficient feedstock to operate new processing capacities [34]. This in turn requires reliable bush biomass stock assessments, which remain limited outside Namibia [35].

To map existing biomass resources, this study draws on three assessments of encroacher bush distribution in Botswana, Namibia, and South Africa [36–38]. While the Namibian and South African studies are relatively recent (2019 and 2020), the Botswana assessment relies on 2002 data. In addition, the studies employ differing methodologies, which complicates their direct comparison and practical integration.

While this study focusses on encroacher bush biomass, the approach is also applicable to invasive alien species that proliferate in new places and create novel ecosystems, such as those in South Africa [23].

2.3. Research gap and scope of work

To date, no rural electrification study has evaluated encroacher bush or alien invasive plants as biomass resources. This study explores three core questions:

- 1) Under which circumstances are encroacher bush based HRES economically feasible – based on the system's LCOE?
- 2) Which tiers of electricity access can the population afford through the most feasible system layouts – based on their WTP?
- 3) How does biomass utilisation contribute to bush encroachment mitigation – based on utilised mass and sustainably cleared land?

In addressing these questions, the study makes three key contributions: (i) it extends techno-economic comparisons of SG-based HRES by modelling realistic, randomised grid unreliability; (ii) it introduces a novel survey-based WTP framework capturing affordability and electricity access across different wealth levels, applying the World Bank's

Multi-Tier Framework (MTF) [39]; and (iii) it links biomass utilisation to bush-encroachment mitigation potential.

3. HyRECA Framework

HyRECA, developed collaboratively by universities, research institutes, private enterprises, and community representatives from Southern Africa and Europe, integrates expertise in renewable energy, climate adaptation, and community-led development. It demonstrates how decentralised HRES based on encroacher bush biomass can provide reliable, affordable, and inclusive energy solutions for rural communities.

3.1. Technical system design

The HyRECA system integrates AgriPV with a baseload-capable SG, comprising a controlled autothermic downdraft gasifier fuelled by encroacher bush and a generator converting syngas into electricity. This pairing ensures reliability, eliminates diesel use and supports constant high loads. A BESS balances generation and demand, providing rapid response to load fluctuations. Electricity is distributed via (i) a mini-grid, (ii) portable batteries for remote users, and (iii) optional grid feed-in. Integrated productive-use components meet domestic and welfare needs while serving as deferrable loads, enhancing efficiency and resilience. Fig. 1 illustrates the system configuration.

3.2. Socio-economic framework

HyRECA systems are deeply integrated in communal water-energy-food systems and act as social innovation hubs for involving citizens in income-generating activities. They rely on upstream biomass supply value chains (harvesting, logistics, preparation) and enable downstream agricultural, artisanal, and industrial uses of electricity services. Kick-starting a virtuous cycle, this income perspective boosts purchasing power, making electricity access affordable and improving living standards.

HyRECA leans on community-owned cooperatives structured as Distributed Energy Service Companies (DESCOs) functioning as inclusive, commercially viable utilities and delivering productive energy and local jobs. These cooperatives manage operations and maintenance (O&M), energy services, and related value chains, supported by skills development initiatives. Community organisations and locally driven independent power producers are key to meeting local needs while creating a sense of ownership in the community [5].

The DESCO model spans full up- and downstream value chains, from biomass logistics and power plant operations to water supply and agricultural activities. Reliable electricity access also attracts third-party businesses as DESCO customers. For grid-adjacent communities, surplus electricity sales may become a significant income source, especially considering ongoing market liberalisation in Southern Africa.

4. Methodology

The following section outlines the methodology, describes the case study site, and details input parameters and assumptions for the HOMER Pro® simulation, impact assessment and upscaling approach.

4.1. Simulation and scenario definition

To evaluate the proposed HRES, this study applies the methodological framework outlined in Fig. 2. HOMER Pro® (v3.18.4) conducts a combinatorial optimisation within a defined search space, where SG, PV, DG, and BESS capacities are systematically varied in increments of 20 kW or kWh, from 0–100 kW (SG & DG), 0–140 kW (PV), and 0–180 kWh (BESS), while grid capacity is optimised up to a maximum of 999,999 kW. All feasible configurations are assessed to minimise total

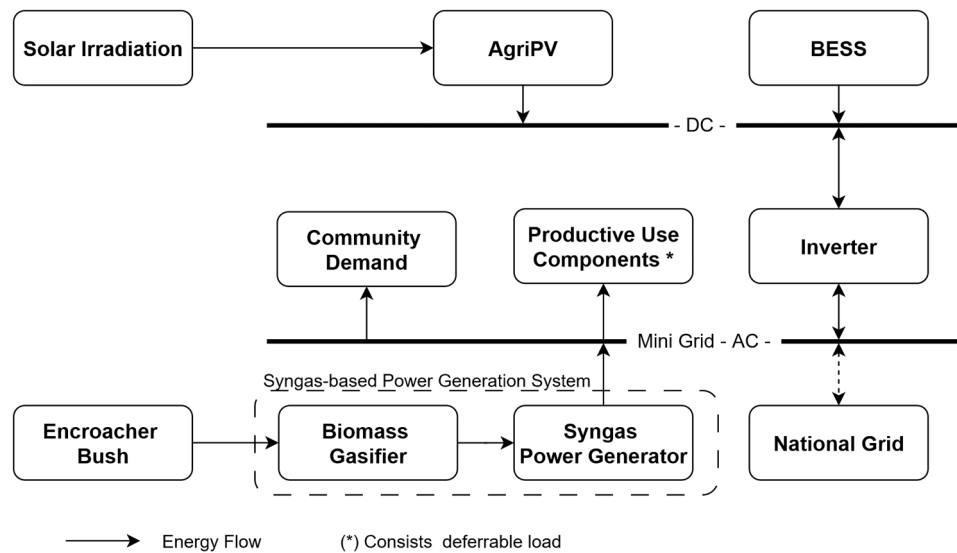


Fig. 1. Schematic diagram showing overall system layout of the HRES.

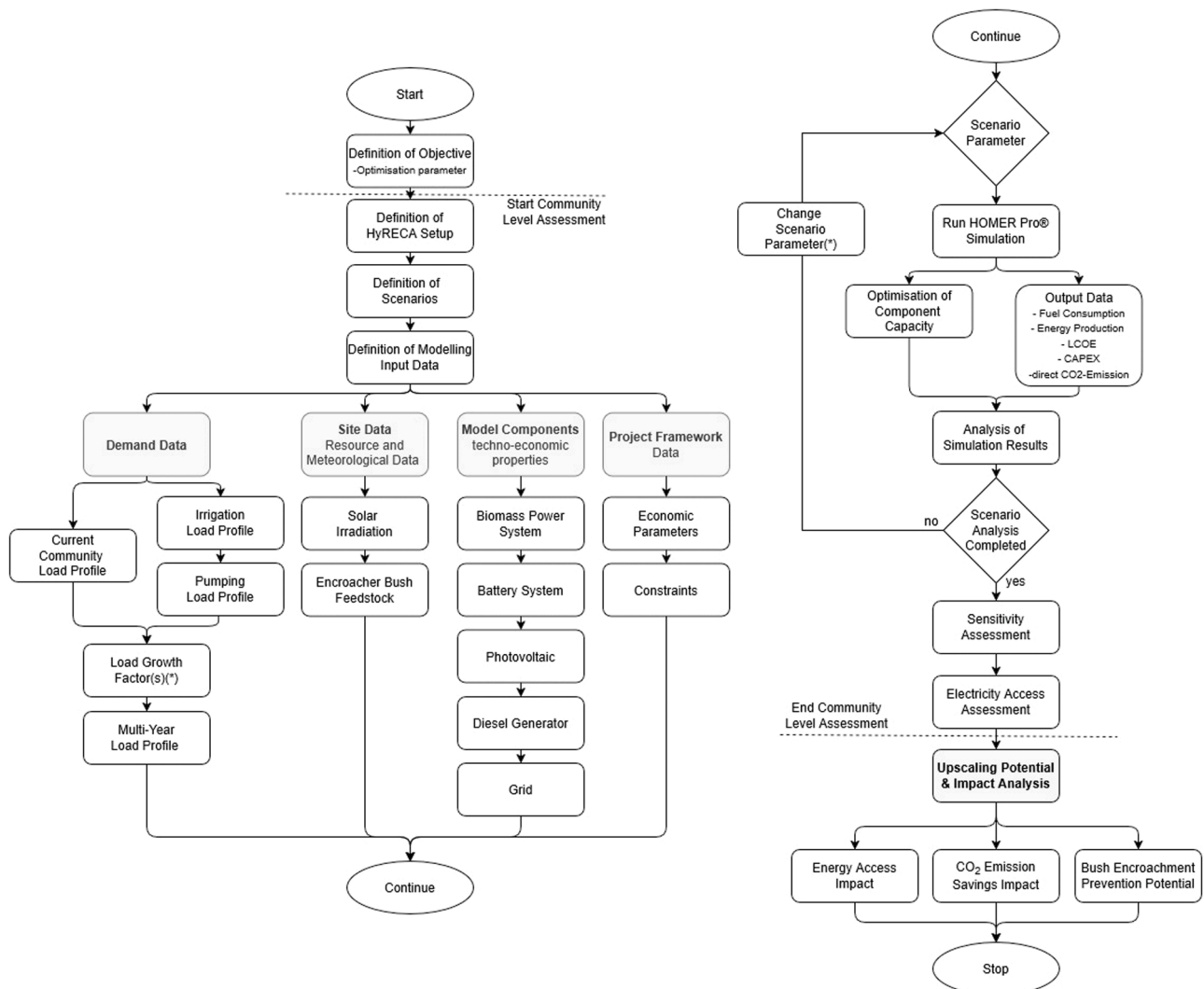


Fig. 2. Flowchart illustrating the methodological framework for the simulation-based evaluation of HRESs with HOMER Pro® on community level.

system cost under a 0.0 % annual capacity shortage constraint, ensuring maximum reliability – contrasting Botswana's frequent grid outages.

Operating reserve is set at 10 % of the current load to cover short-term demand spikes and 80 % of solar PV output to mitigate sudden drops in generation. The simulation assumes a 7.5 % discount rate [40] and 2.0% inflation.

A 25-year project lifetime with hourly time steps is simulated to capture seasonal variability and long-term demand development. To address demand uncertainty, three load growth scenarios – **pessimistic (0 % p. yr.)**, **realistic (+20 % p. yr.)**, and **optimistic (+40 % p. yr.)** – are applied in HOMER Pro®'s multi-year analysis over the first five years post-installation, reflecting varying socio-economic development trajectories as recommended by GIZ's mini-grid sizing guide [41]. Two additional scenarios support grid analysis: Scenario 4 lifts the 0.0 % capacity shortage constraint to simulate current grid conditions, while Scenario 5 introduces a 0.05 \$/kWh feed-in tariff to assess surplus electricity sales. The HOMER Pro® workflow (Fig. 2) includes: (i) defining load profiles (cf. 4.2.2), (ii) integrating meteorological and resource data (cf. 4.2.3), (iii) specifying component parameters (cf. 4.3), and (iv) setting economic constraints. After finalising inputs and system architecture, the optimisation generates outputs including fuel use, energy production, LCOE, CAPEX, and CO₂ emissions. It runs for all five scenarios, followed by a sensitivity analysis of the realistic case and an electricity access assessment based on the optimal system layout. An upscaling potential analysis then extends the site-specific HOMER Pro® results using the optimal off-grid system configuration (cf. 4.6).

4.2. Case study site and demand assessment

Jamataka is a low-income rural village in central Botswana with about 1,125 residents, located 45 km west of Francistown (21.08° S, 27.14° E; WGS 84) [42]. It comprises 156 private households [43] and key public facilities including the Kgotla (traditional meeting place), administration offices, a primary school, and a health clinic (Fig. 3). Due to its dispersed layout and low population density (~50 people/km²) [44], the HRES is centrally located near public facilities and supplies them via a mini-grid, while households outside the mini-grid area access electricity through a centralised portable batteries rental scheme (cf. 3.1). Jamataka lies along the A30 highway, providing access to over 80 km of encroacher bush for a steady biomass supply (Fig. 3). Its proximity to the national grid also enables testing feed-in options to evaluate financial feasibility.

4.2.1. Stakeholder engagement

Stakeholder engagement in Jamataka has been central to designing the energy system. It began with a community energy needs study [43], informing a DESCO model operated by the Jamataka Multipurpose Co-operative Society Limited (JMS) to embed local ownership and decision-making. Workshops with the village chief and Village Development Committee aligned the system with community priorities in agriculture, small businesses, and services. Upstream, bush biomass use creates further opportunities – harvesting, logistics, and supply are managed by JMS, linking electricity access with economic development. Ongoing input from Robert Hanna, founding director at African Sun Energy (Pty) Ltd and JMS industry partner has supported this study.

4.2.2. Load profiles

Three load profiles – community, irrigation, and deferrable pumping – were defined and validated through a pre-project workshop with local partners (cf. 4.2.1). Remaining uncertainties are addressed via three load growth scenarios (cf. 4.1). Rural national electricity consumption statistics for validation purposes are not available, which is why the community load profile is cross-checked against Samikannu et al. (2022) who report a demand of 165 kWh/day for Jamataka – deeming this updated assessment plausible [45]. More information is found in supplementary material S3.

4.2.2.1. Community Load Profile. The community load profile for Jamataka is based on village-specific survey data and a daily profile from Motladiile et al. (2019), extrapolated to 2022 using recent population figures [46,47] (Table 1). A seasonal profile is generated based on national statistics [48], with HOMER Pro® variability settings (10 % day-to-day, 20 % timestep) applied. Initial demand averages 226.9 kWh/day with an average load of 9.45 kW and a peak of 34.5 kW. Fig. 4 illustrates daily (1), seasonal (2), and yearly profiles (3).

4.2.2.2. Irrigation. Irrigation demand is modelled using NREL's Rural African Load Profile Tool (ID 79, Version 01-04-2018) [49], which provides a typical daily pattern with a midday dip and a 1,170 W peak. The profile is seasonalised using average monthly rainfall data for central Botswana [50,51], assuming two maize cultivation cycles per year. Demand is set at 100 % in the dry season (May–August) and 10–25 % in the rainy season (September–April), with random variability applied (10 % day-to-day, 20 % timestep). Average demand is 4.46 kWh/day across all scenarios (Fig. 5).

4.2.2.3. Water Pumping. Water is pumped from boreholes into a storage tank, allowing operation during high energy availability and improving system efficiency. HOMER Pro® treats pumping as a deferrable load, prioritising household and irrigation needs but shifting to primary when the tank is empty. Potable water demand – 20.6 litres per capita daily [52] – is linked to population growth and reflected in the three load growth scenarios. The pessimistic scenario assumes no increase, while realistic and optimistic scenarios scale demand annually using Botswana's population growth rate [47] over a five-year period (cf. 4.1). Average pumping demand is 7.85 kWh/day, with a tank storage equivalent to 17.19 kWh.

4.2.3. Energy resources

Solar global horizontal radiation data for Jamataka is obtained through HOMER Pro®, using monthly averages from NASA's Prediction of Worldwide Energy Resources (POWER) database (Jul 1983–Jun 2005), downloaded on August 11, 2025. Daily radiation ranges from 4.51 kWh/m²/day in June to 6.60 kWh/m²/day in January [53] (Fig. 6).

Encroacher bush biomass is collected from roadsides (cf. 4.2), with *Acacia tortilis* analysed as the dominant local species. A fresh sample harvested near Jamataka underwent proximate and ultimate lab analyses, confirming suitability for gasification due to favourable energetic properties and low moisture content after 4–6 weeks of sun-drying (cf. supplementary material S4). Similar properties in other encroacher bush species confirm broader energy potential [54].

Biomass input data is shown in Table 2. In HOMER Pro®, monthly availability is set to 9,999 t/d (“unlimited”) to avoid constraining the optimisation process. Actual availability is assessed post-simulation by comparing modelled HRES consumption with estimated long-term harvestable biomass along 80 km of accessible roadside – based on bush density and regrowth rate values from Namibian experience validated through a local bush biomass assessment by our industry partner (cf. 4.2.1). Seasonal variation is assumed minor, with chipped bush material stored near the SG to ensure year-round supply.

4.3. HRES component characteristics

This study applies literature-based assumptions validated through stakeholder engagement (cf. 4.2.1). Table 3 summarises the key techno-economic parameters, derived from literature and recent market investigations to ensure scientific validity and practical relevance.

The bush SG is modelled using the methodology by Ribo-Perez et al. (2021) for simulating gasifiers in HOMER [55]. As HOMER Pro® cannot model thermochemical reactions, the gasifier is represented by using a static conversion model using a fixed gasification ratio of 1.85 kg/kg to estimate syngas output from bush biomass input (cf. 4.2.3). The



Fig. 3. HyRECA distribution concept and an example roadside with and without bush encroachment.

Table 1
Aggregate energy demand per day for Jamataka community based on adjusted data from [46].

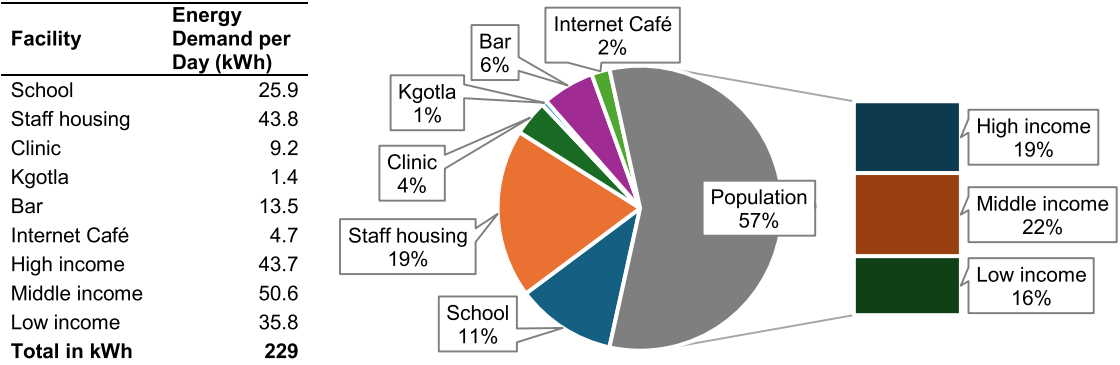


Fig. 4. Community load profile of Jamataka village; derived from HOMER Pro® (1) Daily, (2) Seasonal, (3) Yearly.

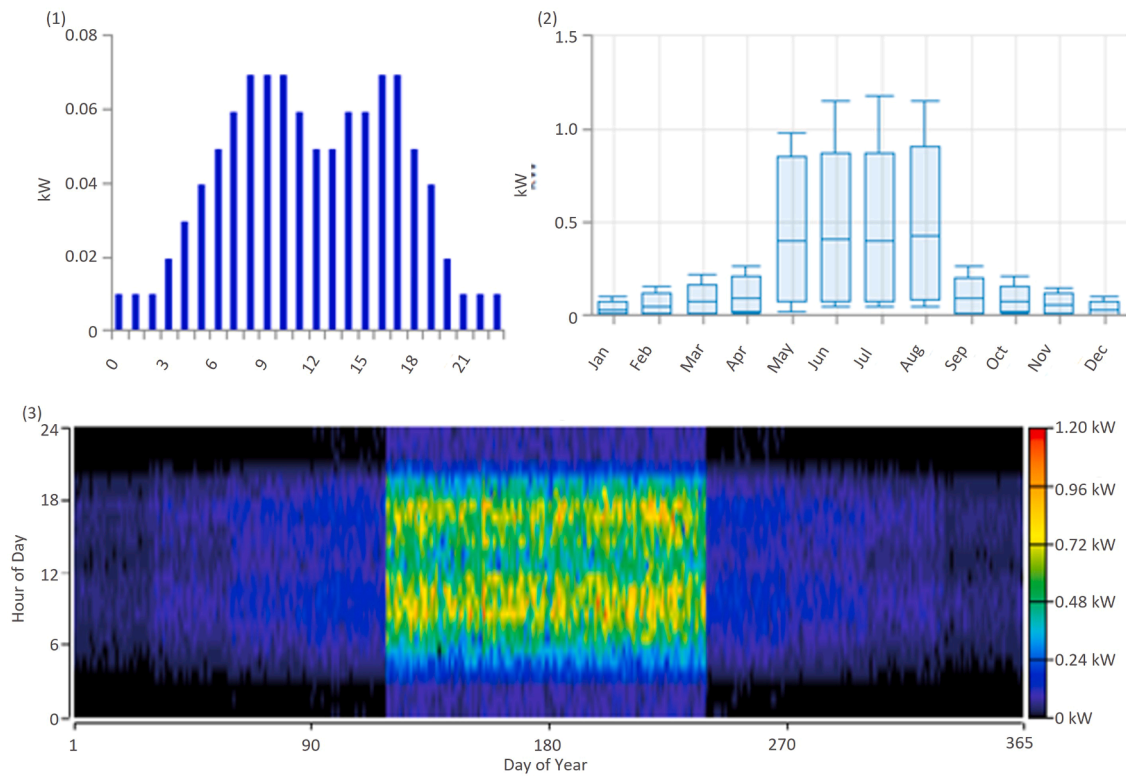


Fig. 5. Irrigation load profile of Jamataka village; derived from HOMER Pro® (1) Daily, (2) Seasonal, (3) Yearly.

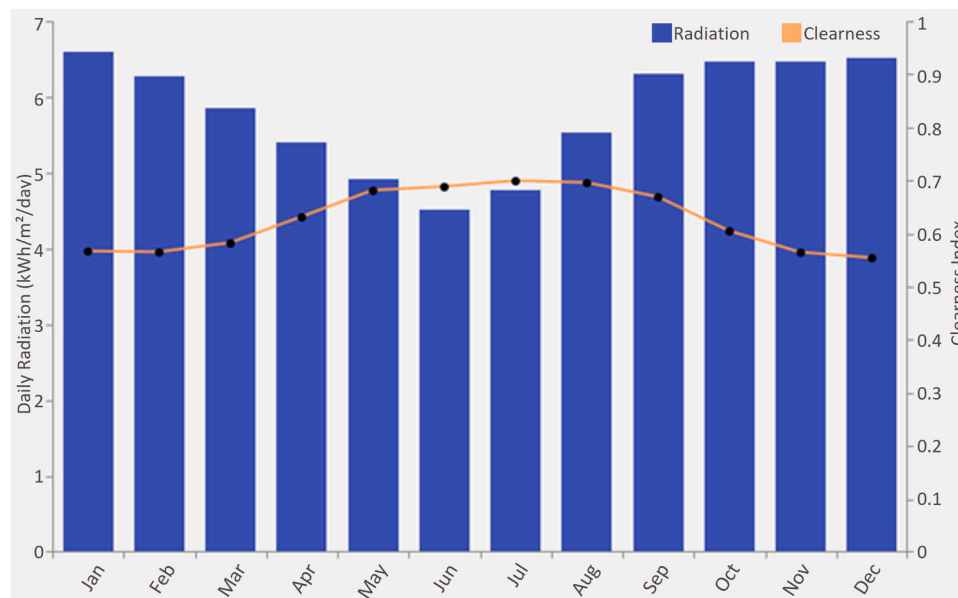


Fig. 6. Global horizontal radiation and clearness index for Jamataka; derived from HOMER Pro®; (cf. supplementary material S4).

generator is adapted from a generic HOMER Pro® unit to reflect syngas operation, based on manufacturer data for the “Power Pallet – PP30” by All Power Labs (2023) [57]. The PV and BESS are modelled using default technical parameters, with cost data obtained through personal communication with our industry partner in Botswana¹. Although the case study considers AgriPV, this does not affect energy modelling,

¹ personal communication with Robert Hanna, founding director at African Sun Energy (Pty) Ltd

which focuses on sizing and system interactions. A DG is included as a benchmark and supplementary power source, represented by HOMER Pro®’s generic diesel generator component.

Grid purchase capacity is set as unlimited, with Botswana’s grid carbon intensity at 1,348 g-CO₂/kWh [60]. Surplus electricity sales are excluded in the three main scenarios and considered only in the dedicated sellback case (cf. 4.1). Typical grid outages last 2–6 hours, occurring almost weekly in parts of 2025, while events like equipment theft can extend repairs to days¹. Based on this variability, repair time standard deviation is estimated at ~155 % of the mean outage duration,

Table 2

Biomass resource input data in HOMER Pro®.

Ref.	Parameter	Value
Ass.	Available biomass	9,999 t/d
Ass.	Average price	30 \$/t
lab data	Carbon content	0.2975
[55]	Gasification ratio	1.85 kg/kg
[55]	LHV of syngas	2.11 kWh/kg
[56]	Bush density	10 t/ha
[56]	Regrowth rate	10 yrs

Abbreviations: Assumption (Ass.), lab data: derived from own proximate and ultimate lab analyses

Table 3

HRES component input data in HOMER Pro®.

Ref.	Parameter	Value
SG System		
[55]	CAPEX	2,000 \$/kWh
[55]	Replacement Cost	1,800 \$/kWh
[55]	O&M	0.072 \$/op. hr
[55]	Lifetime	15,000 op. hr
[55]	Minimum load ratio	0.1
[57]	Minimum runtime	30 min
[57]	Reference generator capacity	20 kW
Der.	Intercept coefficient	0.214 kg/(hr ² kW _{rated})
Der.	Slope	2.143 kg/(hr ² kW _{output})
[57]	Electrical efficiency (Syngas)	0.28
PV System		
1	CAPEX	790 \$/kW
1	Replacement Cost	590 \$/kW
[40]	O&M	9 \$/(kW*yr)
Def.	Lifetime	25 yr.
[45]	Derating Factor	0.88
[45]	Temp. effects on power	-0.4 %/°C
[45]	Nominal operating cell temp.	45°C
Def.	Efficiency (STC)	0.168
DG System		
[58]	CAPEX	800 \$/kW
[58]	Replacement Cost	800 \$/kW
Def.	O&M	0.03 \$/op. hr
Def.	Lifetime	15,000 op. hr
Def.	Minimum load ratio	0.25
Def.	Intercept coefficient	0.033 kg/(hr ² kW _{rated})
Def.	Slope	0.273 kg/(hr ² kW _{output})
Def.	Electrical efficiency	0.33
[59]	Diesel fuel price	1.1 \$/L
BESS (Li-Ion)		
1	CAPEX	300 \$/kW
1	Replacement Cost	300 \$/kW
Def.	O&M	10 \$/(kW*yr.)
Def.	Lifetime	15 yr
Def.	Throughput	3,000 kWh
Def.	Minimum state of charge	0.2
Grid		
Ass.	Purchase capacity	unlimited
Ass.	Sale capacity	none
[60]	Carbon dioxide emissions	1,348 g/kWh
[61]	Grid electricity price	0.11 \$/kWh
Grid reliability		
1	Mean outage frequency	52/yr.
1	Mean repair time	4 hr.
1	Repair time variability	1,55

Abbreviations: Assumption (Ass.), HOMER Pro® Default Value (Def.), Derived from given information (Der.), ¹ personal communication with Robert Hanna, founding director at African Sun Energy (Pty) Ltd

reflecting significant fluctuations in grid availability.

4.4. Impact assessment

The impact assessment evaluates affordability, bush encroachment mitigation and operational CO₂ emissions using the methods described below.

4.4.1. Affordability and electricity access

Affordability is assessed using household WTP survey data for electricity in Jamataka from Mutubuki-Makuyana et al. (2020) [43]. Twelve WTP levels are linked to household groups, and each value is divided by the system's LCOE to estimate monthly purchasable electricity (kWh). These quantities are then mapped to Multi-Tier Framework (MTF) access tiers (Table 4) [39]. Aggregating households by WTP level provides the population distribution across tiers, reflecting affordability-driven electricity access in Jamataka.

4.4.2. Bush encroachment mitigation potential

HyRECA's bush encroachment mitigation potential is assessed using (i) annual mass of encroacher bush utilised ($m_{bush, yr}$, t/yr) and (ii) sustainably managed land area ($A_{bush, total}$, km²). The assessment applies a bush density of $D_{bush} = 10\text{t/ha}$ [56], the calorific value of syngas from *Acacia tortilis* (LHV_{SG} , cf. 4.2.3) and a regrowth rate of $r_{regrowth} = 10\text{ yrs}$ [56] – relying on continuous harvesting rather than full eradication. $m_{bush, yr}$ is derived from HOMER Pro® outputs, linking system demand with conversion efficiencies and syngas LHV (Eq. 1). Total biomass (Eq. 2) and cleared area (Eq. 3) follow, with roadside length l_{road} estimated from biomass availability (Eqs. 4–5), assuming a roadside width of $w_{roadside} = 15\text{ m}$. Given bush biomass is invasive yet limited, its availability is analysed post-simulation to ensure sustainability (cf. 4.2.3).

$$m_{bush, yr} = \sum_{t=1}^{8760} E_{el,t} \cdot \eta_{el, SG} \cdot LHV_{SG} \cdot \eta_{GR} \quad (1)$$

$$m_{bush, total} = \sum_{t=1}^{25} m_{bush, yr, t} \quad (2)$$

$$A_{bush, total} = \sum_{t=1}^{25} \frac{m_{bush, yr, t}}{D_{bush} \cdot r_{regrowth}} \quad (3)$$

$$d_{bush, road} = \frac{D_{bush}}{r_{regrowth}} \cdot 2 \cdot w_{roadside} \quad (4)$$

$$l_{road} = m_{bush, total} / d_{bush, road} \quad (5)$$

4.4.3. CO₂ emissions

HOMER Pro® estimates operational CO₂ emissions from fuel carbon content, assuming that all carbon not emitted as carbon monoxide or unburned hydrocarbons is released as CO₂. For grid-connected systems, CO₂ emissions equal net grid purchases multiplied by the corresponding emission factor, yielding negative values if exports exceed imports [62]. In HyRECA, sustainably harvested encroacher bush biomass results in near-zero net CO₂-emissions as the bush would otherwise be burned during roadside clearing [63,64].

4.5. Sensitivity analysis

A sensitivity analysis addresses uncertainty in key parameters –

Table 4

Power capacity ratings and household electricity services per tier of electricity access [39].

Electricity access tier	Power capacity rating (kWh/d)	Household electricity services
Tier 1	>0.012	Task lighting and phone charging (≥4 h/day)
Tier 2	>0.2	General lighting and phone charging and television and fan
Tier 3	>1.0	Tier 2 and any medium-power appliances
Tier 4	>3.4	Tier 3 and any high-power appliances
Tier 5	>8.2	Full urban-level service (≥23 h/day)

discount rate, diesel and biomass fuel prices, and SG CAPEX/replacement costs (linked) – with input values summarised in Table 5. Simulations in HOMER Pro® use the realistic load scenario, comparing off-grid and grid-connected systems via LCOE. Demand uncertainty is addressed through the three scenarios (cf. 4.1).

4.6. Transnational upscaling

HyRECA's transnational upscaling in Southern Africa illustrates the potential scale of regional impacts rather than precise predictions. It combines spatial biomass availability with settlement distribution to identify communities where deployment may be feasible under simplified assumptions. Bush encroached areas are mapped from literature-derived shapefiles and overlaid with OpenStreetMap settlement data for Botswana, Namibia, and South Africa [36–38,65], enabling identification of settlements within these zones. Suitability for HyRECA deployment is based on population size: settlements with 900–10,000 inhabitants are considered feasible for HRES, while larger settlements are assumed to be grid-connected, and smaller settlements are assumed to have insufficient load for economic HRES operation. Village level population data exist only for Botswana [42]. For Namibia and South Africa, settlement populations are estimated using Botswana's median village size (1,099 inhabitants). Settlements within national parks are excluded to avoid conflicts with conservation policies.

Upscaling applies Jamataka case study results to all eligible settlements identified across the three countries by means of linear extrapolation, assuming (i) similar rural energy demand and biomass availability; (ii) broadly comparable bush density and regrowth; (iii) Botswana's median village population as a proxy for Namibia and South Africa; and (iv) implementation only in non-protected areas. This provides indicative estimates of potential electricity access, biomass utilisation, and CO₂ emission reductions, illustrating HyRECA's potential regional impact under these assumptions.

5. Results and discussion

The upcoming section provides an analysis and discussion of the results from the HOMER Pro® simulation and impact assessment, followed by a discussion of study limitations and recommendations.

5.1. Scenario analysis

For each of the three analysed scenarios, the most cost-effective PV/DG, PV/DG/SG, and PV/SG system combinations are identified for both off-grid and grid-connected setups (Table 6). Grid-connected systems are consistently cheaper – due to the low utility price of 0.11 \$/kWh – but have higher CO₂ emissions, lower renewable fractions, and require additional PV and backup capacity to avoid shortages.

5.1.1. Realistic load forecast

The realistic load scenario assumes 20 % annual demand growth over the first five years and serves as the study's reference for sensitivity and grid analyses. **The optimal off-grid system includes 80 kW PV (84.5 % of production), 40 kW SG (15.5 %), and 180 kWh BESS, using cycle charging at an LCOE of 0.246 \$/kWh.** This setup achieves

Table 5
Sensitivity inputs into HOMER Pro®.

Biomass price (\$/t)	Diesel fuel price (\$/L)	Nominal discount rate (%)	SG CAPEX & replacement cost (factor)
10	0.8	3.5	0.5
20	0.95	5.5	0.75
30	1.1	7.5	1.0
40	1.25	9.5	1.25
50	1.4	11.5	1.5

100 % renewable fraction and zero net CO₂ emissions, with annual biomass use (120 t) well below the sustainable threshold (240 t) and 4,000 m of roadside cleared per year. PV supplies most energy (148,978 kWh/y_{r2025}), while the SG covers morning (6–9 AM) and evening (7–11 PM) peaks in years 5–25 (Fig. 7), producing 101,796 kWh/yr₂₀₂₅ at a mean output of 39.4 kW, typically running at full capacity and charging the BESS when needed. This outcome highlights the strong economic and environmental case for decentralised hybrid systems where solar and biomass resources are available, but a grid-connection is not. However, the need for substantial PV and BESS capacity underscores the high upfront CAPEX – a challenge for low-income communities without external financing.

Fig. 8 illustrates two typical operational modes of the optimal off-grid system. In Type (1), on high solar irradiance days (e.g., January 3–5), the BESS drops to 20 % in the morning, triggering the SG to meet peak demand and recharge the BESS. Solar PV then takes over, recharging the BESS as needed. The evening peak is covered by the BESS until it again reaches 20 %, prompting SG operation. Late night and early morning loads are met by the BESS until the next morning peak. Type (2) shows high PV yield (e.g., February 8), where the BESS never drops below 20 % and the SG is not needed. On overcast days (February 9–10), low PV output requires extended SG operation, with the BESS supplying non-peak periods. Deferrable loads are always served during midday PV peaks.

Runner-up off-grid systems replace the 40 kW SG with a 40 kW DG (7.18 % of production) and increase PV to 120 kW (92.8 %) raising LCOE to 0.332 \$/kWh, mainly due to higher diesel fuel costs, partially offset by more PV. The DG operates similarly to the SG but supplies less electricity (17,279 kWh) and often at partial load (avg. 31.4 kW) thanks to higher PV penetration. Switching from biomass to diesel notably increases cost and emissions, underscoring the sensitivity of energy security and affordability to bush biomass availability in rural off-grid communities. DG systems are not viable where bush biomass exists, though sustained reliance on biomass requires consistent bush-clearing and reliable supply logistics – demanding greater land-use coordination and labour than diesel. Operational uncertainties, including logistics, maintenance, and training a skilled workforce to operate a SG system, need to be addressed by the DESCO operating the HRES.

For grid-connected systems, low grid electricity prices make SG options less competitive. **The optimal configuration includes 40 kW PV (58.9 % of production), a 40 kW DG (2.37 %), and grid purchases (38.8 %), using load following at an LCOE of 0.117 \$/kWh.** However, this setup has a low renewable fraction (31.7 %) and high CO₂ emissions (~164 t/year). The DG mainly serves as backup during grid outages, while PV reduces diesel use by supplying power during daytime outages. Fig. 9 shows the overlap between DG output and grid outages.

Fig. 10 shows two operational modes of the optimal grid-connected system. On days with full grid availability (e.g., February 5), grid electricity is used except during high PV yield hours, when PV meets all demand. On overcast days, PV is supplemented by grid power. During grid outages (e.g., February 6–7), the DG supplies nighttime and peak loads, while PV covers daytime demand, allowing the generator to run at minimum capacity. The generator stops once the grid returns. Annually, 54 outages result in 269 hours of DG operation.

The runner-up grid-connected system is the PV/DG/SG combination, with 20 kW each for SG and DGs sharing backup duties. The DG runs primarily, with the SG supporting peak-hour outages. A 40 kW SG (LCOE: 0.131 \$/kWh) is less competitive due to low runtime as grid backup. High-CAPEX SGs need longer annual operation to justify costs, whereas lower-CAPEX diesel systems suit short-term backup despite higher fuel costs. Sufficient PV capacity ensures on-site use of cheap solar power. Overall, grid-connected systems are more cost-effective given low tariffs, but their higher emissions and limited renewable share challenge long-term sustainability – reflecting a broader policy tension in Southern Africa: cheap, fossil-based grid power versus cleaner decentralised renewables at higher short-term cost.

Table 6

Most feasible HRES layouts for PV/DG, PV/DG/SG and PV/SG after optimisation for rural electrification in Jamataka, Botswana; CO₂-emissions based on year 25 in the HOMER Pro® Multi-Year analysis.

Scenario	Grid	PV (kW)	DG (kW)	SG (kW)	BESS (kWh)	Dispatch	LCOE (\$/kWh)	Renewable Fraction (%)	CO ₂ (kg/yr)	Biomass fuel consumption (t/yr)	Length of cleared roadside (m/yr)	Sustainable regrowth
Realistic load forecast	✓	40	40			LF	0.117	31.7	163,985			
	✓	40	20	20		LF	0.120	33.3	161,482	2.62	87	✓
	✓	40		40		CC	0.131	34.2	160,016	5.11	170	✓
		80		40	180	CC	0.246	100	0	120	4,000	✓
		120	40	40	180	LF	0.260	92.5	15,292	53.1	1,770	✓
Pessimistic load forecast		120	40		180	CC	0.332	50.1	74,302			
	✓	20	20			LF	0.116	32.8	78,288			
	✓	20		20		CC	0.128	35.2	76,306	2.69	90	✓
	✓	20	20	20		LF	0.140	33.9	77,408	1.38	46	✓
		60		20	160	LF	0.204	100	0	22.7	757	✓
Optimistic load forecast		60	20	20	160	LF	0.212	98.6	1,303	21	700	✓
		80	40		160	LF	0.258	88.6	10,228			
	✓	60	60		20	CC	0.117	30.1	309,579			
	✓	60	40	20	20	CC	0.117	31.4	305,916	3.84	128	✓
	✓	60		60	20	CC	0.128	32.3	303,093	8.37	279	✓
		100		80	180	CC	0.253	100	0	249	8,300	✗
		120	20	80	120	CC	0.260	86.8	30,441	189	6,300	✓
		140	80		180	CC	0.351	39.8	165,201			

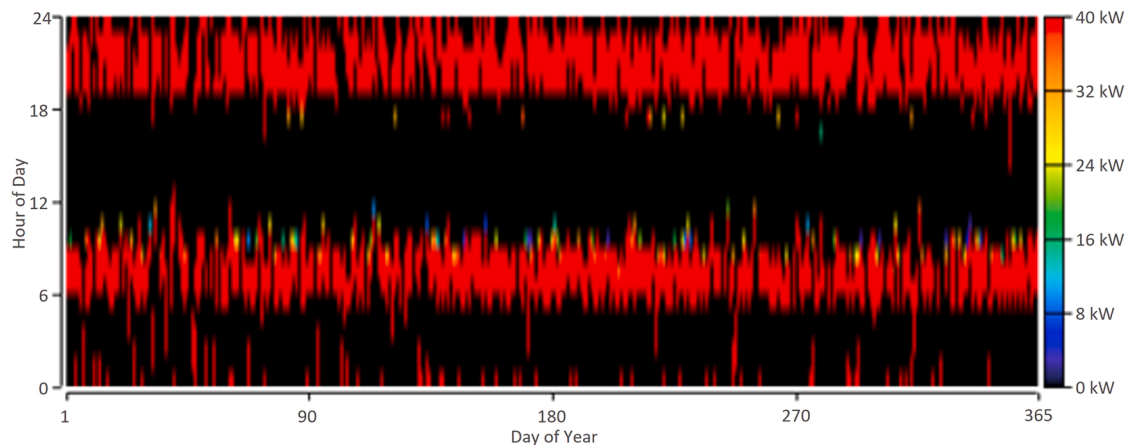


Fig. 7. SG power output; based on year 25 in the HOMER Pro® Multi-Year analysis.

5.1.2. Pessimistic load forecast

The pessimistic scenario assumes zero demand growth. The optimal off-grid system includes 60 kW PV (86.6 % of production), 20 kW SG (13.4 %), and a 160 kWh BESS, with an LCOE of 0.204 \$/kWh. This 100 % renewable system uses 22.7 t/year of biomass, enabling 757 m of sustainable road clearing. Compared to the realistic scenario, component sizes are smaller, and the system follows a load-following strategy with PV charging the BESS and the SG meeting peaks. The grid-connected system uses the same components as in the realistic scenario but at lower capacities: PV (20 kW at 38.8 % of production), DG (20 kW; 2.11 %), and grid purchases (59.1 %). It follows a load-following dispatch strategy with a slightly lower LCOE of 0.116 \$/kWh. CO₂ emissions (~78 t/yr) are reduced due to lower demand, while the renewable fraction remains similar at 32.8 %. The zero-growth scenario represents a cautious electrification pathway: lower capacities limit economic activity – constraining community development and delaying cost recovery, even if the system is affordable upfront. Such systems may serve as scalable starter solutions for early electrification, though this scenario is unlikely for Jamataka.

5.1.3. Optimistic load forecast

The optimistic scenario assumes a 40 % demand increase over the first five years. The optimal off-grid system includes PV (100 kW at 87 % of production), SG (80 kW; 13 %), BESS (180 kWh), and uses cycle charging at an LCOE of 0.253 \$/kWh – slightly higher than in the realistic scenario. It achieves a 100 % renewable fraction and zero net CO₂ emissions, but biomass demand (249 t/yr) slightly exceeds the sustainable limit (240 t/yr), requiring additional clearing. This marks the upper limit of bush biomass-based SGs, making resource management critical for long-term viability and emphasising the need to integrate energy modelling with land-use and biomass regeneration studies.

The grid-connected system mirrors the realistic scenario: PV (60 kW at 70.1 % of production), DG (60 kW; 0.791 %), grid purchases (29.1 %) – plus a BESS (20 kW). It employs cycle charging with an unchanged LCOE of 0.117 \$/kWh. Due to higher demand, CO₂ emissions rise to ~310 t/yr, while the renewable fraction slightly drops to 30.1 %. The BESS is rarely used – mainly during grid outages in morning and night hours – and complements the DG, with a low throughput of 429 kWh/yr. Other components behave similarly, with the DG running at full load

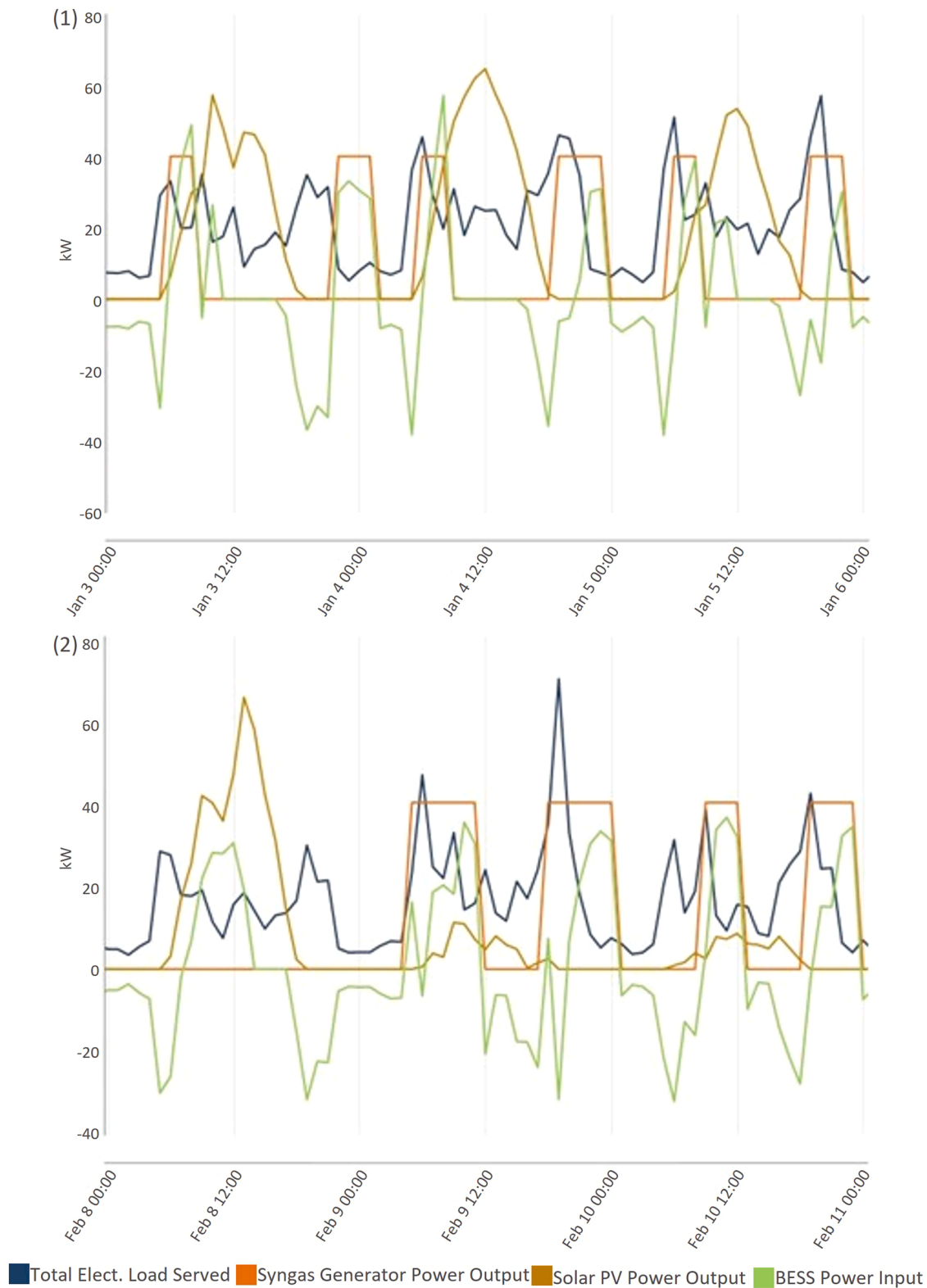


Fig. 8. Total electrical load served and power sources for three days: (1) in January with typical solar PV yields and (2) in February with overcast sky and low solar PV yields; based on year 25 in the HOMER Pro® Multi-Year analysis.

when needed and charging the BESS, consistent with cycle charging behaviour. The unchanged LCOE despite increased demand indicates that economies of scale could offset higher CAPEX, suggesting potential cost reductions as rural demand grows – provided local supply chains and maintenance capabilities expand proportionally.

5.2. Sensitivity analysis

The sensitivity of the optimal off-grid system is assessed using the spider plot in Fig. 11. Biomass fuel price has minimal impact on LCOE, while discount rate and SG capital and replacement costs significantly

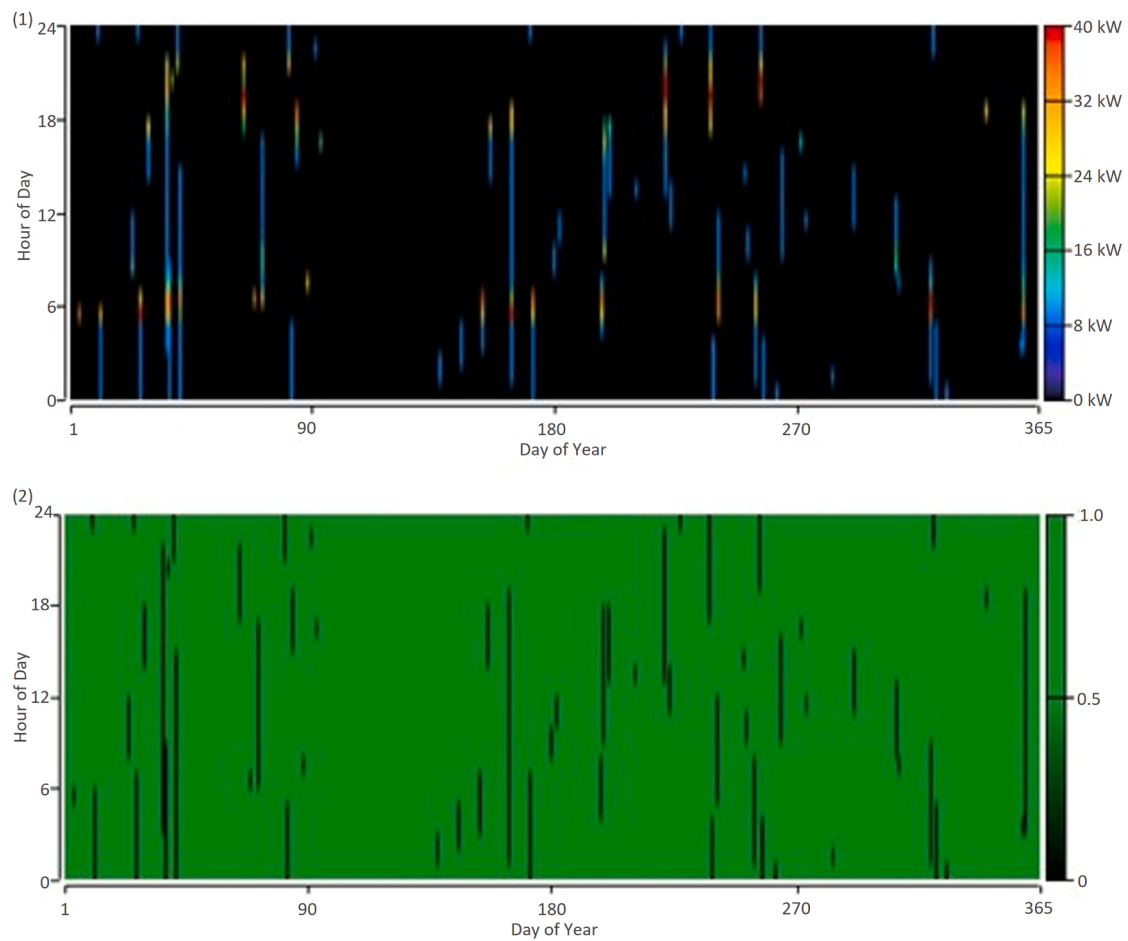


Fig. 9. Overlap of (1) DG power output and (2) grid outages for one year; based on year 25 in the HOMER Pro® Multi-Year analysis.

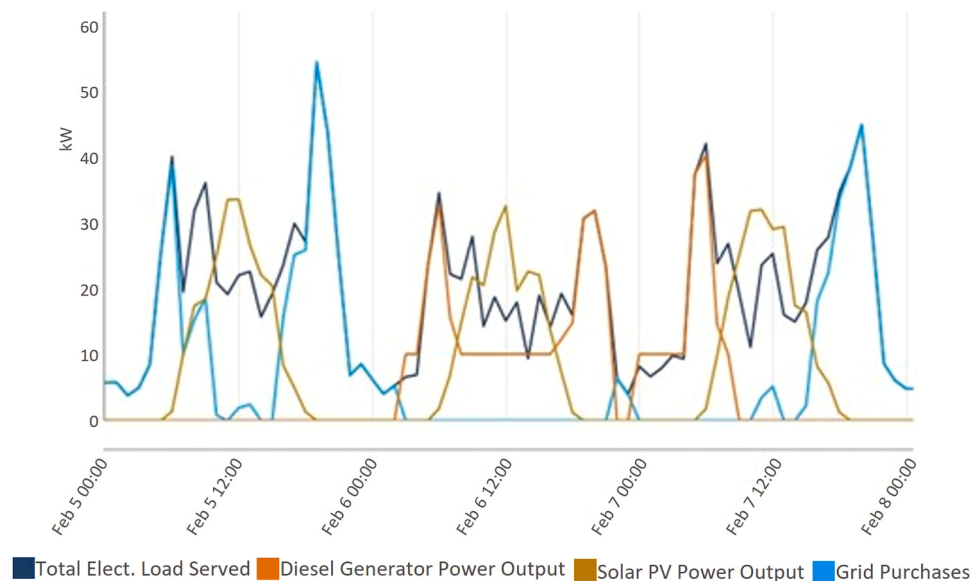


Fig. 10. Total electrical load served and power sources for three days in February including grid power outages – based on year 25 in the HOMER Pro® Multi-Year analysis.

affect results.

The greatest uncertainty lies in SG-related costs, which vary widely in literature—unlike site-dependent diesel fuel and discount rates. These variables are further explored in an optimal system type plot (Fig. 12),

showing which technology mix minimises cost under varying conditions. The SG/PV/BESS system dominates across most scenarios, but at high biomass prices and SG capital & replacement costs above 2,400 \$/kW, the DG/SG/PV/BESS mix becomes more viable. However, high

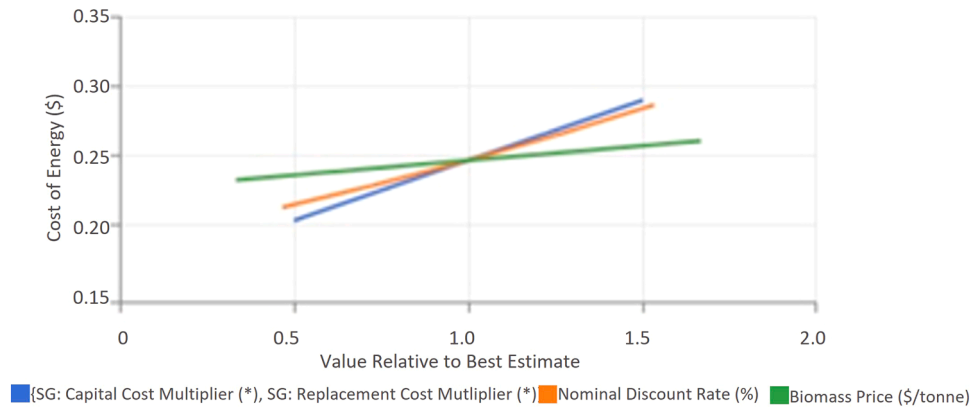


Fig. 11. Spider plot of sensitivity variables for off-grid system.

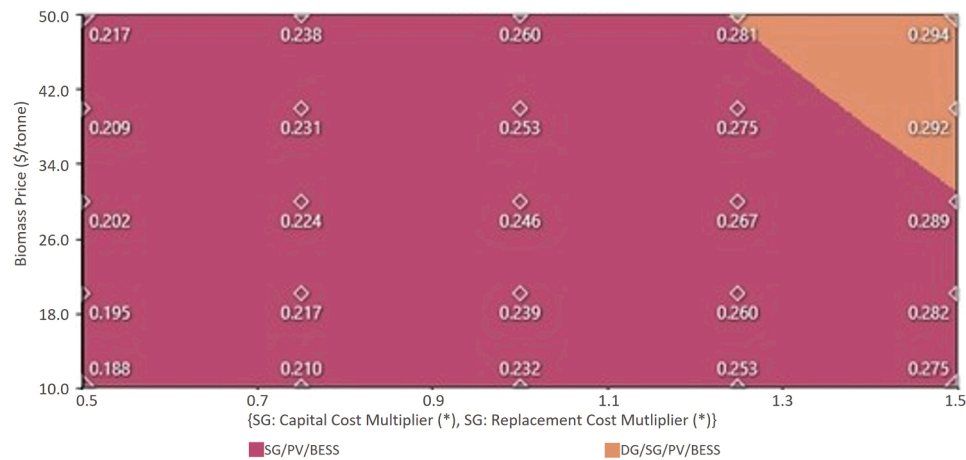


Fig. 12. Optimal system type plot for biomass price and SG capital and replacement cost in an off-grid system; superimposed LCOE (\$/kWh).

biomass prices are unlikely since bush biomass is a waste stream already harvested independently of HyRECA.

The dominance of SG capital and replacement cost in determining system feasibility suggests that policy interventions or R&D aimed at lowering gasifier costs could significantly improve viability. In contrast, the biomass price range tested here may not adequately reflect the variability introduced by differing transport distances or decentralised collection schemes. Future analyses should therefore utilise a more elaborate biomass cost model to assess spatial sensitivity – particularly for sites located further from available biomass resources. The high discount rate sensitivity underscores the importance of financing conditions, as high interest rates can erode the competitiveness of distributed generation.

For the grid-connected optimal system, the spider plot shows minor

sensitivity to diesel price and greater sensitivity to the discount rate (Fig. 13). Overall, variations are smaller than in off-grid systems since most electricity is sourced from the grid.

The optimal system type diagram for grid-connected systems varies diesel price and SG capital/replacement cost multiplier (Fig. 14). Three system combinations fall within the sensitivity range, with DG/PV/Grid dominating. At diesel prices above 1.34 \$/L and SG costs over 1,920 \$/kW, adding a BESS (DG/PV/BESS/Grid) becomes economically viable. The DG/SG/PV/Grid system breaks even at SG costs of 1,680 \$/kW with Botswana's current diesel price of 1.10 \$/L. At the highest tested diesel price (1.40 \$/L), this system nears feasibility at 1,960 \$/kW—just 40 \$/kW below the assumed SG cost of 2,000 \$/kW.

The low overall sensitivity of grid-connected systems reflects their dependence on cheap grid electricity, which insulates them from

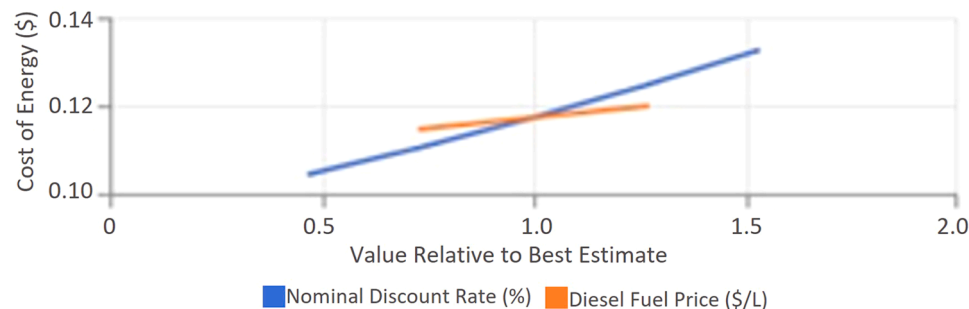


Fig. 13. Spider plot of sensitivity variables for grid system.

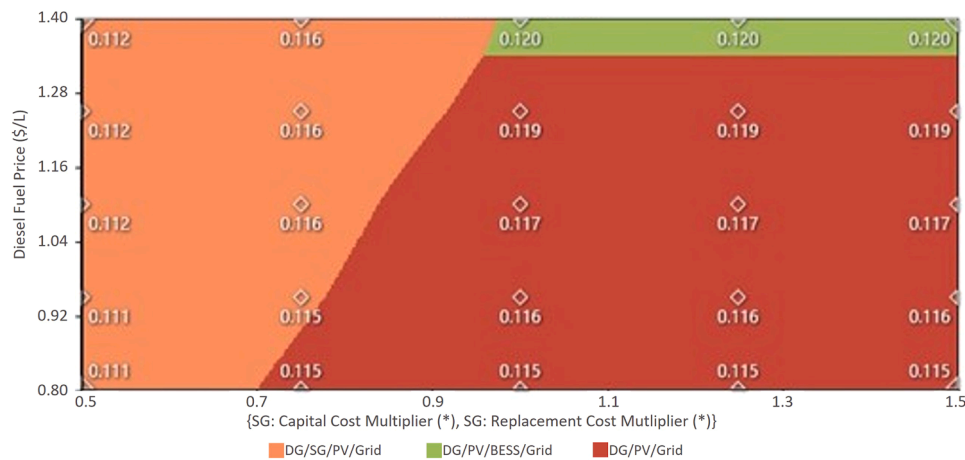


Fig. 14. Optimal system type plot for diesel fuel price and SG capital and replacement cost in a grid connected system; superimposed LCOE (\$/kWh).

fluctuations in fuel or component costs. While biomass gasifiers could become viable with lower CAPEX, carbon pricing, or targeted incentives, their limited operating hours in grid-connected setups restrict economic feasibility. Thus, without policy support or higher fossil fuel costs, conventional DG/PV/Grid systems will remain the least-cost option, despite their continued reliance on imported, carbon-intensive energy.

5.3. Grid system analysis

An extended grid analysis explores the feasibility of grid-connected SG through two scenarios (Table 7). In the first, the zero-capacity-shortage constraint is removed to reflect Jamataka's current grid situation. Simulations show an annual shortage of $>3\%$ and CO_2 emissions of 231 t/yr. Integrating 40 kW of PV reduces both the LCOE ($0.11 \rightarrow 0.0946$ \$/kWh) and capacity shortages ($3.09 \rightarrow 2.47\%$), confirming PV's role as a low-cost reliability enhancer in weak-grid contexts. A small DG eliminates remaining shortages at marginal cost (~ 0.007 \$/kWh), showing limited dispatchable DG is the most cost-effective complement to PV when grid supply is unreliable.

The second scenario introduces a 0.05 \$/kWh sellback price, incentivising 200 kW PV and negative CO_2 emissions via surplus export (per HOMER Pro® accounting). While this reduces LCOE, real-world export constraints (e.g. network limits, curtailment rules) and market access would likely limit such benefits. DG-based grid systems remain cheaper (0.0406 \$/kWh) than SG-based (0.0460 \$/kWh). SG operates only ~ 200 hours per year, consuming ~ 3.9 t biomass – insufficient for a viable bush-clearing value chain, undermining HyRECA's goal of linking electrification with encroacher bush mitigation.

High CAPEX and low runtime render grid-connected SGs uneconomic under current assumptions, even with feed-in tariffs. DG/PV/Grid combinations remain most feasible, while grid-connected SG deployment would require significant cost reductions or targeted policy support – such as carbon pricing, capital subsidies, or differentiated feed-in

tariffs rewarding dispatchable renewable power. Nonetheless, even under the 2030 universal electricity access goal, the government plans to deploy solar off-grid solutions in remote villages beyond the existing distribution network to decrease extension costs [66].

5.4. Literature comparison

The most feasible off-grid HRES yields an LCOE of 0.246 \$/kWh, close to the median value of 0.229 \$₂₀₂₅/kWh obtained from 17 comparable SG-based studies. Consistent with previous work [14–16], the grid-connected HRES achieves a markedly lower LCOE of 0.120 \$/kWh, versus the 0.07 \$₂₀₂₅/kWh median reported across nine grid-integrated studies (cf. 2.1). Differences in LCOE largely stem from CAPEX assumptions. SG CAPEX in the literature range from 400–500 \$/kW [67, 68] to 1,000–1,162 \$/kW [16,69], while this study adopts a more conservative 2,000 \$/kW. As noted by [17], SG systems with high capital but low fuel costs require extended operation to achieve cost-effectiveness, unlike DGs, with low capital but high operating costs.

Table 8 compares this study with two off-grid and two grid-connected cases of similar scale and demand. Discount rates are comparable (5–5.95 %), apart from Rajbongshi et al. (2017), who assume 0 % to reflect government subsidies [15]. All studies, except [70], enforce zero-capacity-shortage. Yimen et al. (2022) employ a configuration similar to this study's off-grid setup but with smaller SG capacity (15 kW vs. 40 kW), driven by their exceptionally high SG CAPEX and replacement costs (6,000 / 5,500 \$/kW), three times higher than applied here. PV and battery costs are also significantly higher (3x and 2x, respectively), yielding an LCOE of 0.352 \$₂₀₂₅/kWh. Conversely, Rajbongshi et al. (2017) assume high PV and battery costs (2,800 \$/kW and 1,295 \$/kW), likely reflecting older technology cost baselines, while their DG CAPEX (296 \$/kW) remains lower than in this study [15]. Biomass fuel costs across all cases fall within 16–30 \$₂₀₂₅/t.

The optimal off-grid system's operational profile resembles that of Yimen et al. (2022), both employing cycle-charging and achieving high

Table 7

Grid system scenario analysis for rural electrification in Jamataka, Botswana; CO_2 -emissions based on year 25 in the HOMER Pro® Multi-Year analysis.

Scenario	Grid	PV (kW)	DG (kW)	SG (kW)	BESS (kWh)	Dispatch	LCOE (\$/kWh)	Annual Capacity shortage (%)	Renewable Fraction (%)	CO_2 (kg/ yr)
Grid - capacity shortage allowed (status quo)	✓					CC	0.1100	3.09	0	231,847
Grid - capacity shortage allowed (cost-optimised)	✓	40				CC	0.0946	2.47	32.9	160,244
Grid - no capacity shortage allowed (cost-optimised)	✓	40	40			LF	0.1170	0	31.7	163,985
Grid - with sellback (PV/DG)	✓	200	40			LF	0.0406	0	79.3	-215,578
Grid - with sellback (PV/SG)	✓	200		40		CC	0.0460	0	80.1	-218,682

Table 8

Comparison of the most feasible HRES layouts with selected optimised layouts from the literature.

Ref.	Year	Location	Peak load (kW)	Energy demand (kWh/d)	PV (kW)	DG (kW)	SG (kW)	BESS (kWh)	Grid	Biomass feedstock	LCOE (\$ ₂₀₂₅ /kWh)
[17]	2019	India	49	200	30		20	120		manure, straw, wood waste	0.275
[70]	2022	Cameroon	56	287	82		15	200		wood waste	0.352
	2025	Botswana	35	227	80		40	180		bush (<i>Acacia tortilis</i>)	0.246
	2025	Botswana	35	227	40	20	20	0	✓	bush (<i>Acacia tortilis</i>)	0.120
[16]	2022	India	29	88	10		5	0	✓	pine needles	0.109
[15]	2017	India	41	286		5	40		✓	-	0.113

PV shares (84–92 %) complemented by SG generation [70]. In Singh et al. (2019), limited solar resources and lower PV capacity result in SG providing about two-thirds of generation [17]. Among grid-connected studies, Malik et al. (2022) report 62 % SG, 20 % PV, and 18 % grid imports for a smaller community, reflecting modest PV yields under Himalayan conditions [16]. By contrast, this study's grid-connected setup is PV-dominant (59 %), with 39 % grid supply and only 2 % DG/SG backup – highlighting Botswana's high solar potential, where PV is primary and SG/DG serve as resilience during grid outages. Three of the four comparative studies analyse biomass fuel cost sensitivity [15, 16,70], finding minimal LCOE impact (<3 %) for price variations of up to ± 40 %, indicating low cost sensitivity [16,70]. Comparable stability is observed in this work.

In summary, the cost advantage of grid-connected over off-grid systems aligns with previous literature, confirming the model's directional validity. Variations mainly reflect regional CAPEX and policy assumptions rather than methodological discrepancies. The close agreement in system performance and sensitivity outcomes with comparable studies demonstrates that the model reliably captures the key techno-economic dynamics of SG-based HRES under Southern African conditions.

5.5. Affordability and electricity access

The affordability analysis for Jamataka applies twelve WTP levels (50–1,000 BWP) from Mutubuki-Makuyana et al. (2020) to estimate the amount of electricity that households can afford based on the LCOEs for, both, grid-connected and off-grid systems (cf. 4.4.1) [43]. Table 9 presents households' distribution across WTP levels and affordability results, while Fig. 15 shows the aggregated share of households per electricity access tier for off-grid and grid-connected systems.

Results indicate that grid-connected systems offer higher electricity access in Jamataka than off-grid solutions. For the off-grid system, households paying <100 BWP remain at Tier 2, while all grid-connected

households reach at least Tier 3. Most households report a WTP of 50 BWP – about what they currently spend on paraffin and candles for lighting [43]. Even at Tier 2, these households gain lighting and small device charging, so they still benefit under the off-grid option. Tier 3 dominates in both systems (>50 %). For grid-connected, Tier 4 is second largest (40.8 %) versus only 10.5 % off-grid. Tier 5 access is rare but about three times more common with grid-connection. Both systems meet MTF reliability and quality requirements under zero-capacity-shortage, with legality and safety ensured via DESCO operation (cf. 3.2).

5.6. Transnational upscaling

OpenStreetMap data show 6,155 settlements in Botswana, Namibia, and South Africa lie in bush-encroached areas [65]. About 81 % are large cities or suburbs, generally outside HyRECA's rural electrification scope (cf. supplementary material S5). Considering villages only, 926 settlements remain, with 244 villages (636,846 people) in Botswana [42], 84 in Namibia (~92,000 people), and 828 in South Africa (~910,000 people) – suggesting that HyRECA could serve ~1.35 million people – about half of the 2.7 million in the region currently lacking electricity access (Fig. 16) [4].

At village scale, HyRECA sustainably utilises ~3 kt of biomass over 25 years, clearing ~3 km² while allowing for regrowth. Extrapolated to eligible settlements, this corresponds to ~2,800 km² and ~2,800 kt of biomass – which is still <1 % of the region's 315,595 km² bush-encroached land. Furthermore, HyRECA systems could potentially avoid ~5.4 Mt CO₂ over 25 years versus Botswana's grid mix and ~1.7 Mt versus PV/DG systems. These figures should be interpreted as indicative, not predictive, as they assume similar settlement structures, biomass availability, and socio-economic conditions across regions – real-world variation may alter impacts.

Overall, this upscaling exercise illustrates HyRECA's regional potential to expand rural electricity access, mitigate bush encroachment,

Table 9

Household's willingness to pay for electricity per month in Jamataka [43], affordable electricity consumption per day and electricity access tier for off-grid and grid-connected systems.

Willingness to pay per month (BWP)	Number of households	Share of households	Off-grid system		Grid-connected system	
			Affordable electricity* (kWh/d)	Electricity access tier	Affordable electricity* (kWh/d)	Electricity access tier
50	17	22.4 %	0.5	2	1.1	3
60	1	1.3 %	0.6	2	1.3	3
80	3	3.9 %	0.8	2	1.7	3
100	13	17.1 %	1.0	3	2.1	3
150	5	6.6 %	1.5	3	3.2	3
200	10	13.2 %	2.0	3	4.2	4
250	9	11.8 %	2.5	3	5.3	4
300	8	10.5 %	3.0	3	6.3	4
350	4	5.3 %	3.5	4	7.4	4
400	2	2.6 %	4.0	4	8.5	5
500	2	2.6 %	5.0	4	10.6	5
1000	2	2.6 %	10.1	5	21.2	5

* Based on the grid-connected LCOE of 0.117 USD/kWh, the off-grid LCOE of 0.246 USD/kWh, a 30-day month, and an exchange rate of 13.47 BWP/USD (24.10.2025)

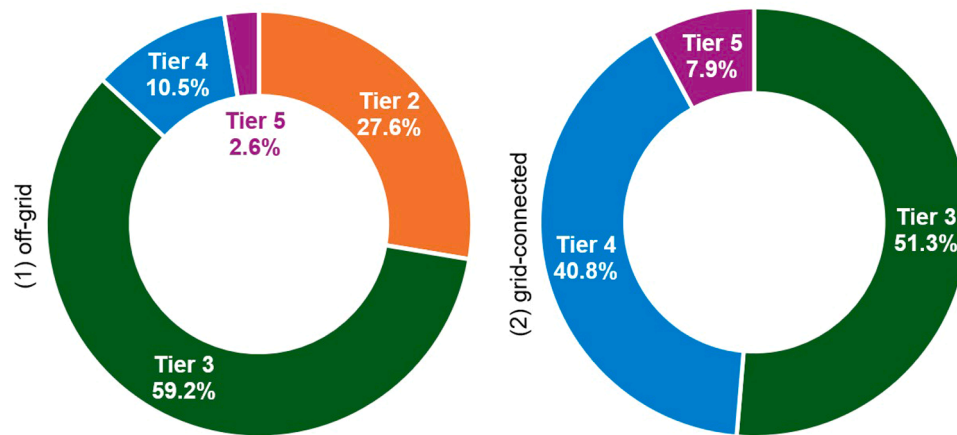


Fig. 15. Share of households per electricity access tier in Jamataka for off-grid (1) and grid-connected (2) system.

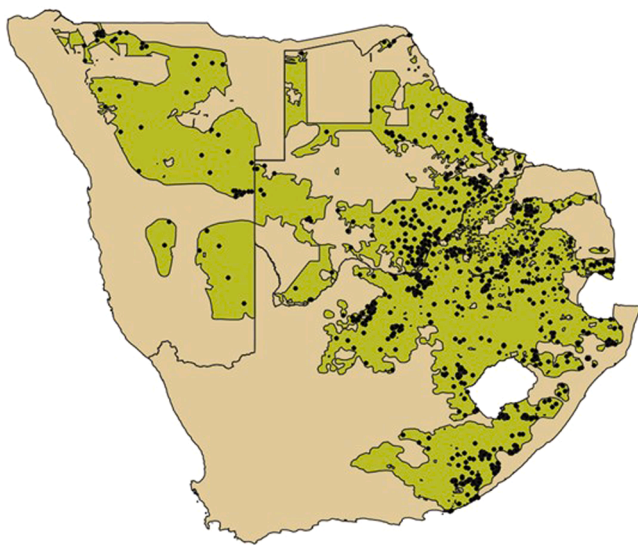


Fig. 16. Settlement locations (black dots) and areas affected by bush encroachment (coloured green) in Botswana, Namibia and South Africa – excluding protected areas and national parks [36–38,65].

and reduce carbon emissions under the outlined assumptions (Table 10).

Table 10

Results of the impact analysis for electricity access, bush encroachment and CO₂-emissions for Jamataka village and Southern Africa including Botswana, Namibia and South Africa (indicative).

	Jamataka, Botswana		Southern Africa	
	per year	25 years (acc.)	per year	25 years (acc.)
Electricity access provided (no. people)	1,125 (total)		~1.35 million (total)	
Bush encroachment				
Cleared bush area (km ² /yr)	0.12	3	~111	~2,800
Cleared bush material (kt/yr)	0.12	3	~111	~2,800
CO₂ emissions (kt)				
PV/SG (HyRECA)	0	0	0	0
...saved compared to PV/DG	0.07	1.9	~69	~1,700
...saved compared to grid (status quo)	0.23	5.8	~215	~5,400

5.7. Limitations and recommendations

Several factors limit the generalisability and precision of this study's results. The modelling based on HOMER Pro®, simplifies real-world system behaviour and may not fully capture load growth, biomass regrowth, grid outages, component degradation, or dispatch strategies under regional conditions. Uncertainty also exists in SG costs and biomass availability, with heterogeneous and partly outdated data emphasising the need for updated, harmonised resource mapping that includes energy densities, growth rates, and site-specific logistics. All HOMER Pro® input data are sourced from literature or local stakeholders, validated through stakeholder engagement, and sensitivity analyses were conducted to assess uncertainty in the most relevant parameters. The emissions assessment is limited to direct CO₂ emissions, with life cycle impacts excluded from the study scope.

This analysis does not account for the high upfront grid-connection fees charged by Botswana's utility company which often prevent households from accessing electricity despite their ability to pay monthly costs. In such cases, decentralised off-grid systems – particularly those using portable batteries – could offer a more affordable and inclusive alternative for low-income households.

This study's transnational upscaling is an illustrative assessment of HyRECA's regional potential. It uses simplifying assumptions, including similar energy demand, biomass availability, bush density, regrowth rates, and Botswana's median village population as a proxy for settlements in Southern Africa. Consequently, estimates of electricity access, biomass utilisation, and CO₂ emission reductions are indicative, reflecting potential scale rather than exact outcomes. Real-world variations in settlement characteristics, local electricity demand, and ecology could lead to significant deviations and may require distinct feasibility studies.

Future research should prioritise updated bush biomass mapping and high-resolution geospatial modelling that integrates data on bush distribution, settlements, infrastructure, and grid proximity, alongside catchment-specific assessments of resource availability, logistics, and electrification options. Developing a biomass cost model to capture spatial price variations is essential. The upscaling approach should then be refined to reflect these improvements for more accurate, locality-specific assessments. Integrating carbon markets [71], techno-economic modelling, and participatory approaches can improve financial feasibility and socio-economic inclusivity. Finally, socio-economic studies on value chain dynamics and community impacts will strengthen system viability and adoption.

Policy frameworks should support SG-based HRES through concessional financing, subsidies, R&D, and local manufacturing to reduce costs and enhance local value. Grid-connected SG deployment

will require significant cost reductions or targeted policy support – such as carbon pricing, capital subsidies, or differentiated feed-in tariffs rewarding dispatchable renewables. Sustainable biomass harvesting via community-based schemes and DESCO oversight should align with bush-encroachment mitigation strategies, ensuring climate benefits through the carbon neutrality of processed biomass. Regionally, harmonised Southern African Development Community (SADC) policies could facilitate knowledge exchange, technology transfer, and scaling of the HyRECA framework beyond single sites.

6. Conclusions

This study explores HyRECA's potential to increase rural electricity access while mitigating bush encroachment in Southern Africa through HRES combining AgriPV and encroacher bush-based SGs. Unlike prior studies, it incorporates randomised grid unreliability into HOMER Pro® LCOE analysis and introduces a WTP framework linking income disparities to World Bank MTF electricity access tiers [39]. By quantifying biomass use, the study offers a holistic perspective on sustainable rural electrification and lays the foundation for future ecological restoration efforts [71].

Scenario and sensitivity analyses demonstrate a clear divergence in economic and environmental performance between off-grid and grid-connected system architectures:

For off-grid communities, PV/SG/BESS systems consistently proved most cost-effective across all demand scenarios, delivering 100 % renewable energy, near zero net CO₂ emissions, and LCOEs of 0.204–0.253 \$/kWh. In the realistic scenario, the optimal mix – PV (80 kW), SG (40 kW), and BESS (180 kWh) with cycle charging – achieves an LCOE of 0.246 \$/kWh while sustainably utilising ~3,000 t of biomass and clearing ~3 km² of bush encroached land over 25 years. Under the optimistic scenario, bush biomass demand slightly exceeds sustainable limits, highlighting the need for careful resource management. DG options were consistently less attractive due to higher LCOEs and emissions. Sensitivity analysis confirms PV/SG/BESS as the dominant configuration, with DG-systems only competitive under unrealistically high SG CAPEX and fuel costs.

For grid-connected communities, low tariffs (0.11 \$/kWh) make PV/DG/Grid combinations most cost-competitive. In the realistic scenario, the optimal mix – PV (40 kW), and a DG (40 kW) with load following – achieves an LCOE of 0.117 \$/kWh but performs poorly environmentally, with low renewable fractions (30–33 %) and high CO₂ emissions (78–310 t/yr). DGs remain the preferred backup during outages due to low upfront costs and suitability for short runtime operation. Even with incentives like feed-in tariffs, SG cannot achieve cost parity at current CAPEX levels – only significant carbon pricing would improve its prospects. Sensitivity analysis confirms this stability: grid-connected HRES economics are insensitive to parameter changes, as most demand is met by the utility grid.

In summary, SG-based HRES are the most viable, sustainable, and cost-effective option for off-grid rural electrification, while diesel backup remains economically dominant in grid-connected contexts despite its environmental drawbacks. The results thus highlight a critical distinction: the case for SG strengthens significantly in off-grid communities, whereas its competitiveness diminishes once access to low-cost grid electricity is available – a finding consistent with the literature and current rural electrification plans in Botswana [66].

Affordability analysis for Jamataka shows grid-connected systems deliver higher electricity access, with all households reaching at least Tier 3, while many low-income households on the off-grid system remain at Tier 2. Still, off-grid options provide meaningful access for lighting and small appliances, improving on resident's current use of paraffin and candles. High upfront grid-connection fees limit practical grid-access, highlighting the potential of decentralised off-grid solutions with portable batteries for low-income households.

A first-order estimate suggests that applying HyRECA across 926

bush-encroached villages in Botswana, Namibia, and South Africa could provide electricity access to over 1.35 million people, sustainably clear ~2,800 km² of bush biomass, and avoid up to 5.4 Mt CO₂ over 25 years – compared to Botswana's grid mix. While this covers <1 % of the encroached area, it underscores both feasibility and the vast untapped potential of encroacher bush biomass as a renewable resource in Southern Africa. Further analysis beyond Jamataka is needed to refine these projections and validate transnational feasibility.

This study demonstrates HyRECA's potential to improve rural electricity access and mitigate bush encroachment in Southern Africa, while highlighting uncertainties around biomass data and upscaling potential. Addressing these challenges requires updated bush resource mapping, high-resolution geospatial modelling, and village-level assessments of demand, logistics, and electrification pathways. Future research should further explore socio-economic impacts and carbon markets integration, while policy support through concessional finance, sustainable biomass governance, and SADC-level coordination could enhance feasibility and scalability. Together, these steps would strengthen the evidence base and enable HyRECA to evolve from a case study into a regionally integrated strategy for sustainable rural electrification.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Microsoft Copilot in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published paper.

CRediT authorship contribution statement

Stuart Daniel James: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Markus Killinger:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Chiedza Ngonidzashe Mutanga:** Writing – review & editing, Writing – original draft. **Romain Pirard:** Writing – review & editing, Writing – original draft. **Mario Einax:** Writing – original draft. **Matthias Huber:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Tobias Bader:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

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Data availability

Data will be made available on request.

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